

ASSIGNMENT – 1

NAME: Manish Madhukar Gawas

Roll no: 3202008

CASE STUDY:

Amazon EC2 (Elastic Compute Cloud) is a key service provided by Amazon Web Services (AWS) that allows users to run virtual servers in the cloud. It is part of AWS's Infrastructure as a Service (IaaS) offering, enabling businesses to scale computing resources quickly and cost-effectively.

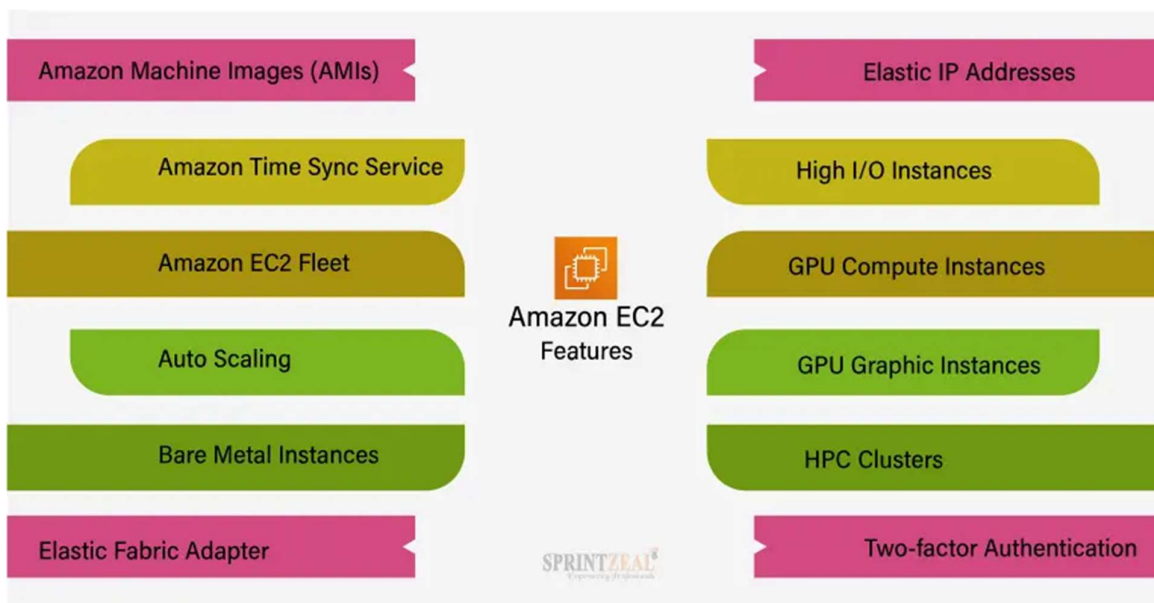
Basic Explanation

1. What is Amazon EC2?

Amazon EC2 allows you to rent virtual servers, known as **instances**, in the cloud. These instances run on physical hardware hosted in Amazon data centers. EC2 gives you flexibility to choose the operating system, instance type, and software to install on your server, depending on your needs.

Key Features:

- **Scalability:** You can easily add or remove instances to adjust to traffic demands.
- **Pay-as-you-go pricing:** You only pay for the compute power you use.
- **Variety of instance types:** Instances range from small, low-cost options to large, high-performance ones.
- **Customization:** You can choose specific configurations, such as operating system (Linux or Windows), CPU power, and storage.



2. How Does Amazon EC2 Work?

1. **Launch an Instance:** You can create an EC2 instance through the AWS Management Console, CLI, or APIs. You specify:

- **Instance Type:** What kind of virtual server (e.g., CPU, memory, storage).
 - **Operating System:** Which OS the instance will run (e.g., Linux, Windows).
 - **Security Group:** Set up firewall rules for your instance.
2. **Access and Use:** After launching the instance, you can SSH into a Linux server or RDP into a Windows server to install software and run applications.
 3. **Scaling:** You can increase or decrease the number of instances based on your needs, and the cloud will automatically manage scaling if configured with Auto Scaling.
 4. **Termination:** When you no longer need the instance, you can terminate it, stopping any further charges.

Detailed Explanation

1. Instance Types

Amazon EC2 provides a variety of instance types optimized for different workloads, including:

- **General Purpose:** Balanced resources for a variety of applications (e.g., T-series, M-series).
- **Compute Optimized:** Ideal for applications requiring high compute power (e.g., C-series).
- **Memory Optimized:** For workloads that need high memory (e.g., R-series).
- **Storage Optimized:** For workloads needing high disk throughput (e.g., I-series, D-series).
- **Accelerated Computing:** For GPU-based tasks (e.g., P-series, G-series).

Each instance type comes with varying amounts of CPU, RAM, and storage configurations.

2. Virtualization

EC2 instances are based on virtualization technology, meaning each instance runs on a physical machine that hosts multiple virtual machines. Amazon uses a hypervisor called **Xen** or **Nitro** to manage virtual instances. These hypervisors enable efficient allocation of resources, isolation between instances, and support for high-performance networking and storage.

3. Elastic Load Balancing (ELB)

When you scale your EC2 instances horizontally (adding more instances to handle increased traffic), you can use **Elastic Load Balancing (ELB)** to distribute traffic evenly across multiple instances. ELB improves the availability and fault tolerance of your application by ensuring that no single instance is overloaded with too much traffic.

4. Auto Scaling

EC2 can be configured to automatically adjust the number of instances based on traffic demands. This is done using **Auto Scaling Groups**, which can:

- Automatically increase the number of instances during peak traffic periods.
- Scale down when traffic is lower, ensuring cost efficiency.

Auto Scaling ensures that the application always has the right amount of resources without over-provisioning.

5. Elastic Block Store (EBS)

While EC2 instances provide compute power, **Amazon EBS (Elastic Block Store)** provides persistent storage. Unlike the temporary storage that comes with EC2 instances (which is lost when the instance is terminated), EBS volumes are persistent and can be detached from one instance and attached to another.

6. Networking and Security

EC2 instances are deployed within **Virtual Private Clouds (VPCs)**. A VPC is a logically isolated section of AWS where you can define your network architecture, including subnets, routing, and IP addressing.

Key networking components include:

- **Security Groups:** Act as a firewall, controlling inbound and outbound traffic for your instances.
- **Network ACLs:** Provide additional network security at the subnet level.
- **Elastic IPs:** Static IP addresses that can be associated with your EC2 instances for consistent external access.

7. Pricing Models

Amazon EC2 offers several pricing models to accommodate different use cases:

- **On-Demand Instances:** Pay for compute capacity by the hour or second without long-term commitments. Best for short-term or unpredictable workloads.
- **Reserved Instances:** Commit to using EC2 for a 1- or 3-year term, and receive a significant discount (up to 75%) in exchange for the commitment. Ideal for steady-state applications.
- **Spot Instances:** Bid for unused EC2 capacity at a lower price, but the instances can be terminated by AWS with little notice if demand for capacity increases. This is good for flexible workloads.

8. Monitoring and Management

AWS provides various tools for monitoring and managing your EC2 instances:

- **Amazon CloudWatch:** Monitors EC2 performance metrics like CPU usage, network activity, and disk I/O.
- **AWS Systems Manager:** Helps automate management tasks like patching and configuration updates.
- **AWS CloudTrail:** Logs and monitors API calls made to EC2 and other AWS services, helping with security auditing and compliance.

9. Use Cases for EC2

1. **Web Hosting:** EC2 can host websites ranging from simple blogs to complex e-commerce platforms.

2. **Data Processing:** EC2 is used to run big data processing workloads, like machine learning models or data analytics pipelines.
3. **Development and Testing:** Developers use EC2 instances to create isolated environments for software development and testing.
4. **Enterprise Applications:** Many enterprises run ERP, CRM, and other large-scale applications on EC2, scaling them based on user demand.

Conclusion

Amazon EC2 is a powerful and flexible service that allows users to rent computing resources in the cloud. With features like scalability, pay-per-use pricing, and various instance types, EC2 can accommodate a wide range of applications—from small web apps to large enterprise systems. The combination of EC2 with other AWS services such as Elastic Load Balancing, Auto Scaling, and EBS helps businesses manage resources effectively while optimizing costs.